

UNIVERSITY OF PADOVA

DEPARTMENT OF ECONOMICS AND MANAGEMENT “MARCO FANNO”

Master Degree in Economics and Finance

TESI DI LAUREA

**SOLAR GRID PARITY IN ITALY: WHERE DO WE
STAND
A DYNAMIC STOCHASTIC MODEL**

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ACRONYMS

PV Photovoltaic

NPV Net Present Value

IRR Internal Rate of Return

LCOE Levelised Cost of Electricity

WACC Weighted Average Cost of Capital

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ABSTRACT

Write a single paragraph summarizing your thesis here: the problem you address, your research question, the method you use, your main finding, and its practical or policy implication. Keep it self-contained – a reader should understand your contribution without reading the rest of the thesis.

Keywords: Keyword One, Keyword Two, Keyword Three, Keyword Four

AUTHENTICITY DECLARATION

Dichiaro di aver preso visione del “Regolamento antiplagio” approvato dal Consiglio del Dipartimento di Scienze Economiche e Aziendali e, consapevole delle conseguenze derivanti da dichiarazioni mendaci, dichiaro che il presente lavoro non è già stato sottoposto, in tutto o in parte, per il conseguimento di un titolo accademico in altre Università italiane o straniere. Dichiaro inoltre che tutte le fonti utilizzate per la realizzazione del presente lavoro, inclusi i materiali digitali, sono state correttamente citate nel corpo del testo e nella sezione ‘Riferimenti bibliografici’.

I hereby declare that I have read and understood the “Anti-plagiarism rules and regulations” approved by the Council of the Department of Economics and Management and I am aware of the consequences of making false statements. I declare that this piece of work has not been previously submitted – either fully or partially – for fulfilling the requirements of an academic degree, whether in Italy or abroad. Furthermore, I declare that the references used for this work – including the digital materials – have been appropriately cited and acknowledged in the text and in the section ‘References’.

Firma (signature)

Energy transition research often distinguishes between *grid parity* – the point at which a technology’s levelised cost matches the prevailing market price – and the point at which a rational, uncertainty-averse investor actually commits capital. Classical investment theory shows that under price and cost uncertainty, delaying an irreversible investment can carry positive option value (Dixit & Pindyck, 1994), so the two thresholds need not coincide. Biondi and Moretto (2015) apply this real-options framework to Italian solar photovoltaic investment specifically, motivating the kind of stochastic-modeling approach demonstrated in Chapter 3 of this template.¹

1.1 Motivation

Use this section to state the real-world problem or gap your thesis addresses, and why it matters to the field (policy, industry, or academic literature).

1.2 Research Question and Objectives

State your research question precisely, then list 2–4 concrete objectives that, taken together, answer it.

1.3 Thesis Structure

Briefly describe what each subsequent chapter covers, e.g. Chapter 2 reviews the relevant literature, Chapter 3 sets out the model, and Chapter 4 presents and discusses the results. Fig. 1.1 summarizes this structure graphically.

¹ As a quick illustration of a footnote containing inline math: if a variable follows a Geometric Brownian Motion $dP_t = \mu P_t dt + \sigma P_t dW_t$, its expected value grows at rate μ while its variance is driven by σ .

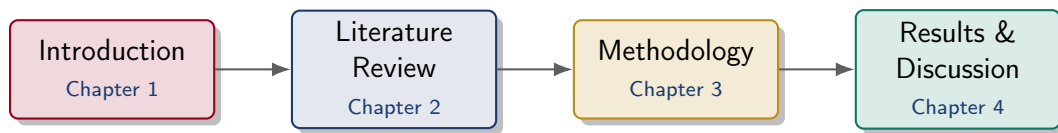


Figure 1.1: Thesis roadmap: how the chapters build on one another.

CHAPTER 2

LITERATURE REVIEW

2.1 Cost and Investment-Timing Studies

Several strands of literature are relevant background for a grid-parity style analysis: cost-projection studies (Lorenczik et al., 2020), learning-curve models of technology cost decline (Breyer & Gerlach, 2013), and stochastic models of electricity price dynamics (Gianfreda & Grossi, 2010; Lucia & Schwartz, 2002). Table 2.1 summarizes a few illustrative examples in the style you might use to structure your own review table.

Table 2.1: Illustrative summary table (booktabs style)

Study	Region	Method	Key Finding
Biondi & Moretto (2015)	Italy	Real options	Investment delay under uncertainty
Lucia & Schwartz (2002)	Nordic	Stochastic price model	Mean-reverting price dynamics
Breyer & Gerlach (2013)	Global	Learning curve	PV cost decline projections

2.2 Positioning of This Thesis

Use this section to state, in a few sentences, what gap in the literature summarized above your own thesis fills.

Table 2.2 shows a second table style – `tabularx` with a fixed total width plus `threeparttable` footnotes – useful when you need to annotate individual values.

Table 2.2: Illustrative parameter table (tabularx + threeparttable style)

Parameter	Description	Value
μ	Drift rate ^a	0.02
σ	Volatility ^b	0.15
r	Discount rate	0.05

^a Annualised, estimated from historical data.

^b Annualised standard deviation of log returns.

3.1 Model Setup

Suppose the relevant state variable P_t (e.g. an electricity price) follows a Geometric Brownian Motion:

$$dP_t = \mu P_t dt + \sigma P_t dW_t, \quad (3.1)$$

where μ is the drift, σ is the volatility, and W_t is a standard Wiener process. Installed system capacity is commonly reported in kW, and irradiance inputs in kW h/m² per year (siunitx handles the unit typesetting and spacing automatically).

Theorem 1: Optimal Investment Trigger

Under a real-options formulation with GBM dynamics as in Eq. (3.1), the optimal investment trigger P^* solves the smooth-pasting condition of the associated Hamilton–Jacobi–Bellman equation. Investment is optimal once $P_t \geq P^*$.

Section 3.1 is an example of the custom theorem box style (built with `\newtcbtheorem`, defined once in `main.tex`) – use it for definitions, propositions, or lemmas that you want visually set off from the main text.

3.2 Solution Approach

Fig. 3.1 sketches the modeling pipeline as a tikz diagram with a locally-scoped style definition (the `\tikzset{...}` call only needs to live in this file; it does not need to be in the preamble). Each stage is colored by role – data, estimation, model, and decision – using the institutional `unipdred` tone from `preamble.tex` alongside three complementary colors (`unipdblue`, `unipdgold`, `unipdgreen`), with a dashed feedback arrow indicating that the trigger is periodically re-estimated as new data arrive.

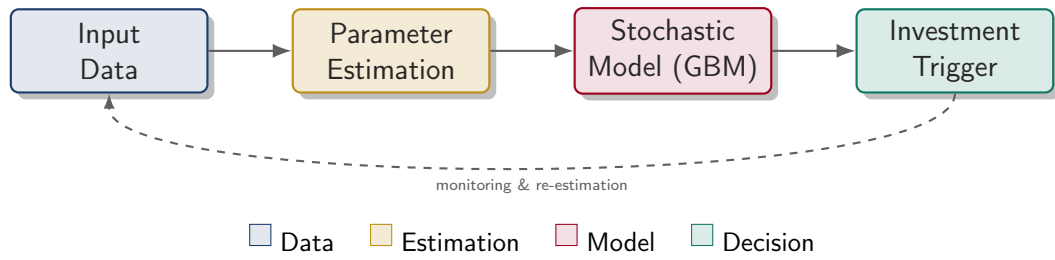


Figure 3.1: Illustrative modeling pipeline: raw inputs are processed into estimated parameters, fed into the stochastic (GBM) model, and used to derive the investment trigger, which is periodically revisited as new data arrive. Colors encode each stage’s role (legend above).

3.2.1 Decision Logic

Fig. 3.2 complements Fig. 3.1 by depicting the real-options decision rule from Section 3.1 as an explicit control-flow diagram: at each period the state variable P_t is compared against the trigger P^* , with investment occurring only once the trigger is reached.

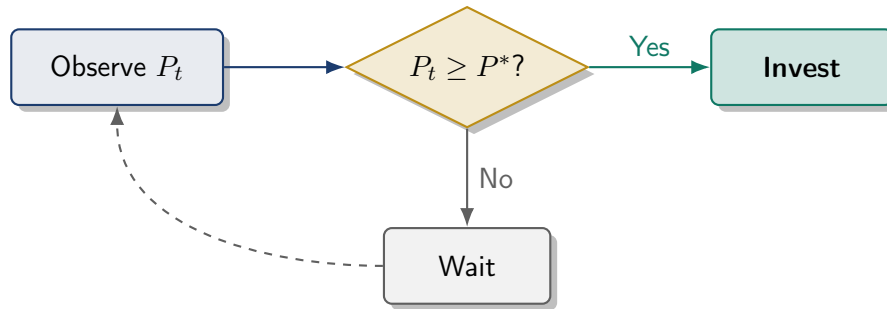


Figure 3.2: Illustrative real-options decision logic: the process observes P_t each period and invests once the trigger P^* is reached; otherwise it waits and re-checks at the next period.

4.1 Illustrative Chart

Fig. 4.1 is drawn directly from inline dummy data with `pgfplots` – useful when you want a chart under version control as text rather than as a binary image file. Swap in your own `coordinates` (or point `pgfplotstable` at a `.csv`) to plot real results.

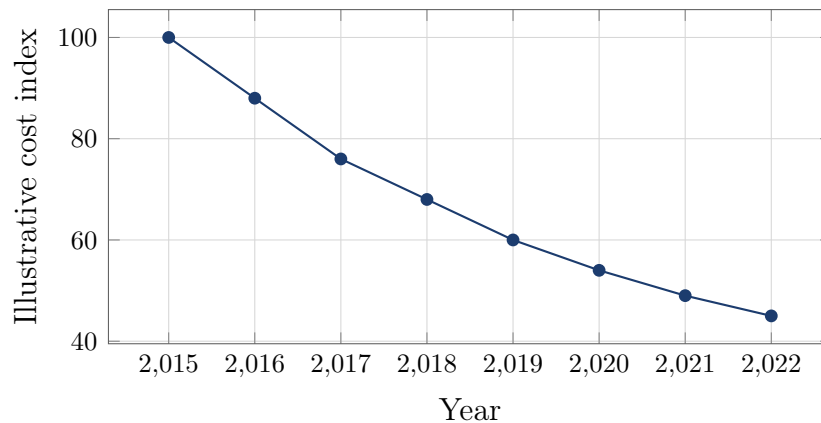


Figure 4.1: Illustrative declining cost index (synthetic data, `pgfplots`)

4.2 Illustrative Code Listing

Fig. 4.2 shows the `minted` package rendering a short script with syntax highlighting. **Note:** `minted` requires Python + `Pygments` installed and `-shell-escape` enabled at compile time – see `README.md` if this block fails to compile.

```
import numpy as np

def simulate_gbm(p0, mu, sigma, steps, dt, seed=0):
    rng = np.random.default_rng(seed)
    path = np.empty(steps)
    path[0] = p0
    for t in range(1, steps):
        shock = rng.normal(0, np.sqrt(dt))
        path[t] = path[t - 1] * (1 + mu * dt + sigma * shock)
    return path
```

Figure 4.2: Illustrative code listing (minted)

4.3 Including an Image File

When you have an actual image file (a plot exported from Python/Stata/Excel, a map, a photo, etc.), include it with `graphicx`'s `\includegraphics`, as shown in Fig. 4.3 (here pointed at the template's own logo file purely to demonstrate the syntax – point it at your own figure).

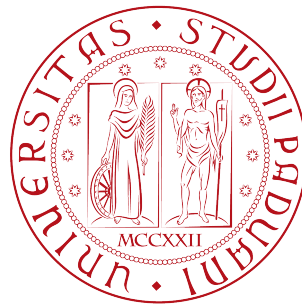


Figure 4.3: Illustrative external image inclusion (`\includegraphics`)

4.4 Discussion

Use this section to interpret your results in light of the literature reviewed in Chapter 2 and the model set out in Chapter 3.

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APPENDIX A

SUPPLEMENTARY DERIVATION

This appendix shows how supplementary material – derivations, robustness checks, extended tables – is structured once `\appendix` has been called in `main.tex`: chapters restart as lettered appendices (A, B, C, ...) instead of numbered chapters.

Theorem 2: Boundary Condition

At the optimal trigger P^* derived in [Chapter 3](#), the value function must satisfy both value-matching and smooth-pasting conditions with the payoff function.

Add your own appendices as additional files in this folder (e.g. `Appendix_B.tex`), and wire each one into `main.tex` with an `\include{appendix/Appendix_B}` line, right after this one.